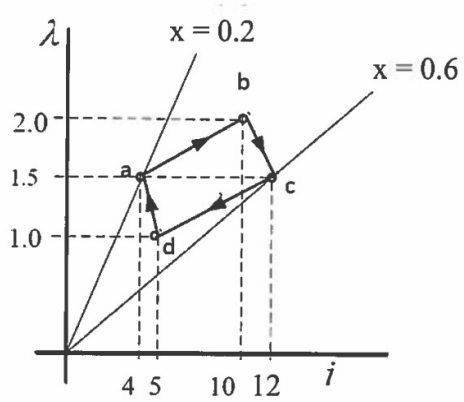
ECE 330

Chapter 5

Analysis of Electromechanical Systems



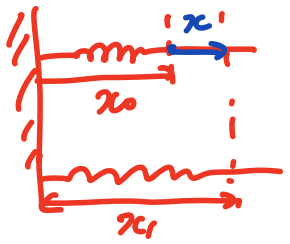
Olga Mironenko
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Fall 2026

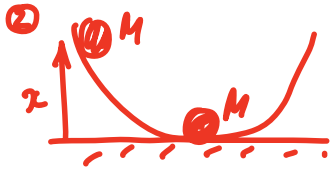
Learning Objectives: you will be able to

- Explain the concepts of **Speed Voltage** and **Transformer Voltage** and calculate them for different electromechanical systems
- Understand the fundamentals of **rotational systems**
- Define and identify **electrically linear systems**
- Calculate **forces of electric origin** via **energy method**
- Define **co-energy**, and calculate **forces of electric origin** via **co-energy method**
- Analyze **energy conversion between two points**
- Define and calculate **EFE** for a system moving between two points
- Define and calculate **EFM** for a system moving between two points
- Define an **energy conversion cycle**, and calculate EFE and EFM over a cycle
- Define a **multiple inputs (ports) system**, and calculate energy, co-energy, and forces/torques for multiport systems

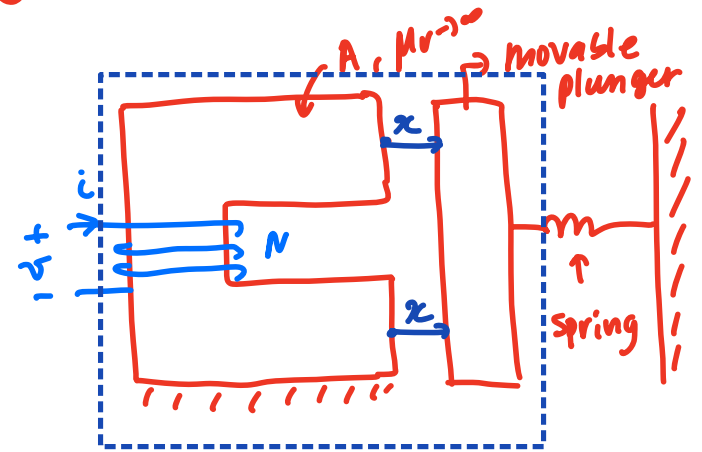
Electromechanical energy conversion: How to make things move using electricity?

* We need to store energy to make things move.

① 

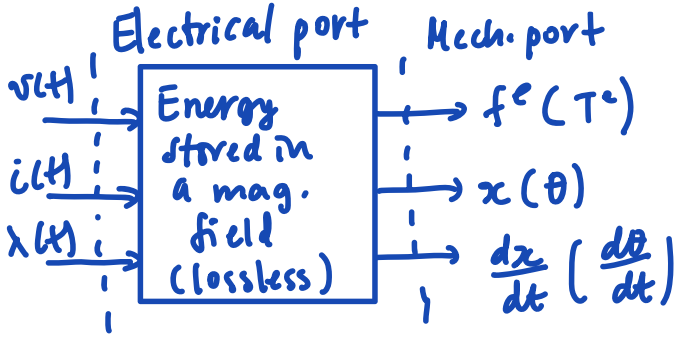
② 

③ Inductors store energy in a mag. field. Where the energy stored? Energy stored in a gap?? energy stored in a core



↑ closed system (lossless, energy is conserved)

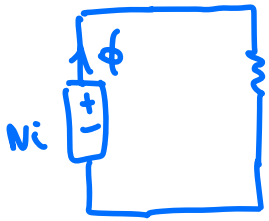
④ Let's abstract general system;



Transformer Voltage and Speed Voltage

4 from electr. port and find $v(t)$:

MEC:



$\mathcal{R}_g(x(t)) \Rightarrow \phi = \dfrac{Ni}{\mathcal{R}_g(x(t))} \Rightarrow \lambda = N\phi = \dfrac{N^2 i}{\mathcal{R}_g(x(t))} \Rightarrow \boxed{\lambda(x,i) = L(x)\cdot i}$

$\mathcal{R}_g(x(t)) = \dfrac{2x(t)}{\mu_0 A}$

$L(x(t))$

Electrically linear

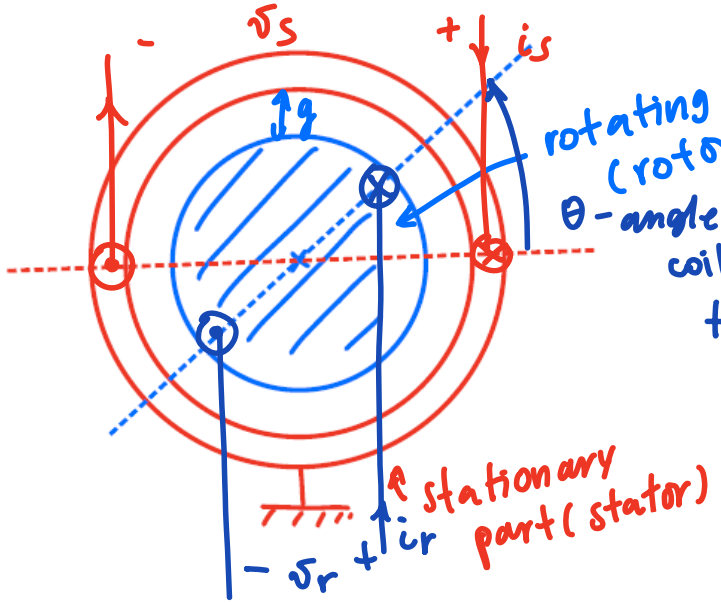
Q: $v(t) - ?$ $\quad v = \dfrac{d\lambda(i,x)}{dt} = \underbrace{\dfrac{\partial\lambda}{\partial i}\dfrac{di}{dt}}_{\text{transformer voltage}} + \underbrace{\dfrac{\partial\lambda}{\partial x}\dfrac{dx}{dt}}_{\text{speed voltage}}$

* Note: If $\lambda(x,i) = L(x)\cdot i$

$$v(t) = L\frac{di}{dt} + i\frac{\partial L}{\partial x}\frac{dx}{dt}$$

* Note: if $x = \text{const}$, $v = L\dfrac{di}{dt}$

Rotational Systems



- $\lambda_s = L_s i_s + \underbrace{(M\cos\theta)}_{L_m(\theta)} i_r$
- $\lambda_r = M\cos\theta\, i_s + L_r i_r$

- $v_s = \dfrac{d\lambda_s(i_s, i_r, \theta)}{dt} =$

$$= \frac{\partial\lambda_s}{\partial i_s}\frac{di_s}{dt} + \frac{\partial\lambda_s}{\partial i_r}\frac{di_r}{dt} + \frac{\partial\lambda_s}{\partial\theta}\frac{d\theta}{dt} =$$

$$= \underbrace{L_s\frac{di_s}{dt} + M\cos\theta\frac{di_r}{dt}}_{\text{transf. voltage}} \underbrace{- M\sin\theta\, i_r\frac{d\theta}{dt}}_{\text{speed voltage}}$$

- $v_r = \dfrac{d\lambda_r(i_s, i_r, \theta)}{dt} = \dfrac{\partial\lambda_r}{\partial i_s}\dfrac{di_s}{dt} + \dfrac{\partial\lambda_r}{\partial i_r}\dfrac{di_r}{dt} + \dfrac{\partial\lambda_r}{\partial\theta}\dfrac{d\theta}{dt} =$

$$= L_r\frac{di_r}{dt} + M\cos\theta\frac{di_s}{dt} - M\sin\theta\cdot i_s\frac{d\theta}{dt}$$

Electrically Linear System: Relating i and λ

Single coil:

$$\lambda(i, x) = L(x) \cdot i$$

* independent variables x and i

$$i(x, \lambda) = \frac{\lambda}{L(x)}$$

* independent variables x and λ

Multiple coils:

$$\lambda_s = L_s i_s + M \cos\theta \, i_r$$

$$\lambda_r = M \cos\theta \, i_s + L_r i_r$$

$$\begin{bmatrix} \lambda_s \\ \lambda_r \end{bmatrix} = \begin{bmatrix} L_s & M\cos\theta \\ M\cos\theta & L_r \end{bmatrix} \begin{bmatrix} i_s \\ i_r \end{bmatrix}$$

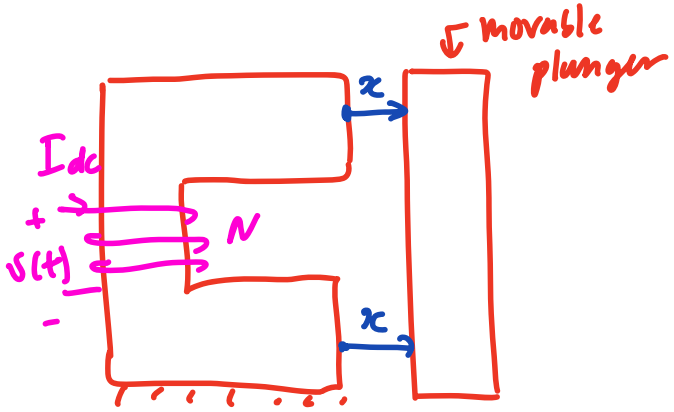
$$\underline{\lambda} = \underline{\underline{L}}(\theta) \cdot \underline{i}$$

$\underline{\lambda}$: flux linkage vector; $\underline{\underline{L}}(\theta)$: inductance matrix (symmetric); $\underline{i}$: current vector

$$\underline{i} = \underline{\underline{L}}^{-1} \cdot \underline{\lambda}$$

$$\begin{bmatrix} i_s \\ i_r \end{bmatrix} = \frac{1}{L_s L_r - M^2\cos^2\theta} \begin{bmatrix} L_r & -M\cos\theta \\ -M\cos\theta & L_s \end{bmatrix} \begin{bmatrix} \lambda_s \\ \lambda_r \end{bmatrix}$$

Example 1



Note:

- Plunger is moving away (x↑)

$$\frac{dx}{dt} > 0$$

- Plunger is moving forward (x↓)

$$\frac{dx}{dt} < 0$$

Given:

Area of the gap,

$A = 10\,cm^2$,

$\mu_{core} = \infty$

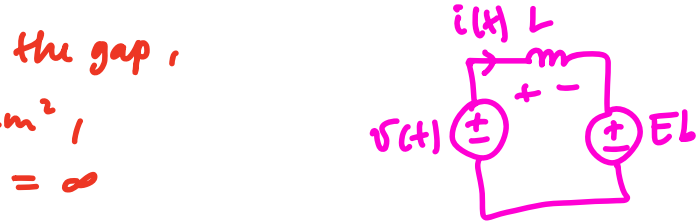
$N = 100$

$I_{dc} = 10\,A$

at $t=0$, gap is forced to open at a rate of 1m/s, and $x(t=0) = 2mm$.

Q: What is the voltage $v(t=0)-$?

① MEC:



$v(t)$ = speed volt. + transf. volt

Note: What does the sign of back emf physically imply?

① $E_b < 0$

Plunger is moving away $\Rightarrow \frac{dx}{dt} > 0$

$\downarrow$

$\mathcal{R} \uparrow \Rightarrow \downarrow \phi = \frac{Ni}{\mathcal{R}\uparrow}$

flux tries to decrease

$\Downarrow$

the back emf must aid the voltage source to increase i to maintain flux (Lenz's law)

② $E_b > 0$

Plunger is moving forward $\Rightarrow \frac{dx}{dt} < 0$

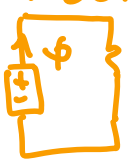
$\downarrow$

$\mathcal{R} \downarrow \Rightarrow \uparrow \phi = \frac{Ni}{\mathcal{R}\downarrow}$

flux tries to increase

$\Downarrow$

back emf opposes the volt. source to maintain flux



$$\mathcal{R}_g = \frac{2x}{\mu_0 A} \Rightarrow \lambda = N\phi = \frac{N^2 i}{\mathcal{R}_g} = \underbrace{\frac{N^2 \mu_0 A}{2x}}_{L(x)} \cdot i$$

el. linear: $\lambda = L(x) \cdot i$

$$② \quad v = \frac{d\lambda(i,x)}{dt} = \frac{\partial \lambda}{\partial i}\frac{di}{dt}^{0\,(I_{dc})} + \frac{\partial \lambda}{\partial x}\frac{dx}{dt} = -\frac{N^2\mu_0 A}{2x^2} \cdot i \left(\frac{dx}{dt}\right) = -15.7\,V$$

$\frac{dx}{dt} = 1\,m/s$

Example 2

Problem 1 (25 points) (Spring 2019, M2, p1(a))

A single-phase generator consists of a coil on the stator and a coil on the rotor with a mutual inductance variation as 0.1cos(θ) (Henries) where θ is the angle from the stator field axis to the rotor field axis. The rotor is being driven at a constant speed of 377 (ω_m) radians per second and the rotor coil has a constant dc current $i_r = 5$ A. The self inductances of the stator and rotor are both constants and you may assume a linear magnetic core.

a) Compute the open-circuit stator voltage ($i_s = 0$) as a function of time (recall that $\int d\theta/dt = \int \omega \Rightarrow \theta = \omega t + \theta_0$ and assume some angle $\theta = \theta_0$ at time zero)

Given: $\lambda_s = L_s i_s + (0.1\cos\theta) i_r$

$\lambda_r = (0.1\cos\theta) i_s + L_r i_r$

Q: $v_s(t)$, open-circuit = ?

$$v_s(t) = \frac{d\lambda_s(i_s, i_r, \theta)}{dt} = \frac{\partial\lambda_s}{\partial i_s}\frac{di_s}{dt} + \frac{\partial\lambda_s}{\partial i_r}\frac{di_r}{dt} + \frac{\partial\lambda_s}{\partial\theta}\frac{d\theta}{dt} = -0.1\sin\theta \cdot i_r \frac{d\theta}{dt} =$$

($i_s = 0$) (0 (i_r is DC); i_r = 5A; $\frac{d\theta}{dt} = \omega_m$ = 377 rad/s)

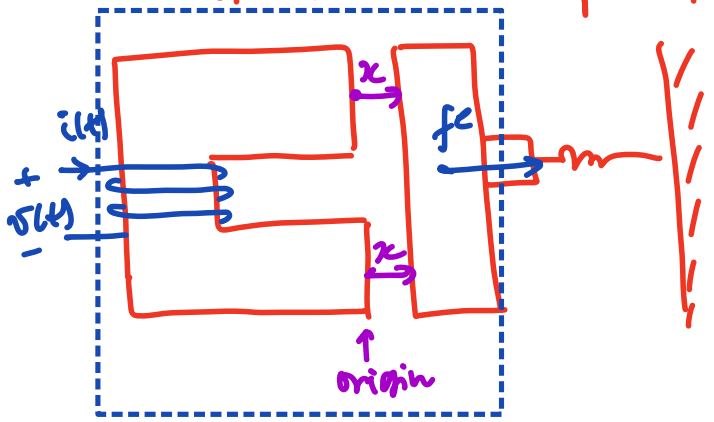
$$= -0.1 \cdot 5 \cdot 377 \sin(\omega_m t + \theta_0) =$$

$$= -188.5 \sin(377t + \theta_0)\ V$$

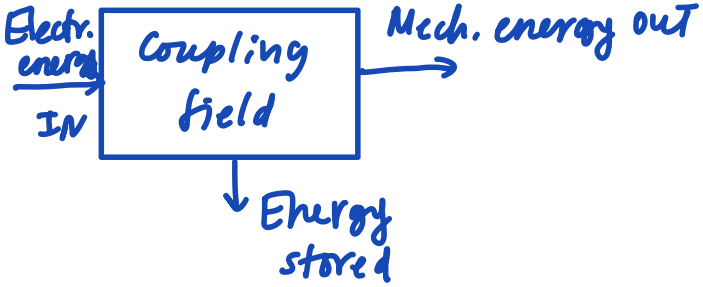
Energy stored in coupling field

Q: f_e at mech. port, calculate the force

↓ force of electric origin f^e: [N]

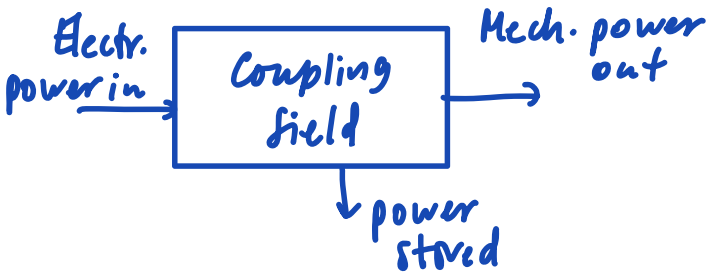


⇒



↑ closed system (energy is conserved, lossless)

OR



* $P_{in} = P_{out} + P_{stored} \Rightarrow P_{stored} = P_{in} - P_{out}$

Mathematical formulation

* $P_{stored} = P_{in} - P_{out}$

* W_m – energy stored in a coupling field [J]

* $\frac{dW_m}{dt} = \underbrace{v \cdot i}_{\frac{d\lambda}{dt}} - f^e \cdot \frac{dx}{dt}$

* $\frac{dW_m}{dt} = i \frac{d\lambda}{dt} - f^e \frac{dx}{dt}$

What is the force of electric origin?

Note: $W_m(\lambda, x) \Rightarrow \lambda, x$ as independent variables

* $\frac{dW_m}{dt} = i\frac{d\lambda}{dt} - f^e\frac{dx}{dt}$

* $\frac{dW_m(\lambda,x)}{dt} = \frac{\partial W_m(\lambda,x)}{\partial \lambda}\frac{d\lambda}{dt} + \frac{\partial W_m(\lambda,x)}{\partial x}\frac{dx}{dt}$

↳ $i = \left.\frac{\partial W_m(\lambda,x)}{\partial \lambda}\right|_{x=\text{const}}$

$$f^e = -\left.\frac{\partial W_m(\lambda,x)}{\partial x}\right|_{\lambda=\text{const}}$$

↑ force is a direction to decrease energy

* Problem: circular definition

Calculating the energy

$$* \int_{t_0}^{t_1} \frac{dW_m}{dt}\,dt = \int_{t_0}^{t_1} i\,\frac{d\lambda}{dt}\,dt - \int_{t_0}^{t_1} f^e\,\frac{dx}{dt}\,dt$$

$$\Downarrow$$

$$\int_{W_{m_0}}^{W_{m_1}} dW_m = \int_{\lambda_0}^{\lambda_1} i\,d\lambda - \int_{x_0}^{x_1} f^e\,dx$$

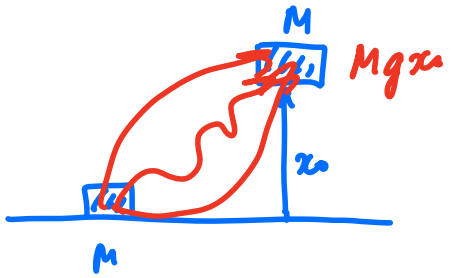
$$\Downarrow$$

$$W_{m_1} - W_{m_0} = \int_{\lambda_0}^{\lambda_1} i\,d\lambda - \int_{x_0}^{x_1} f^e\,dx$$

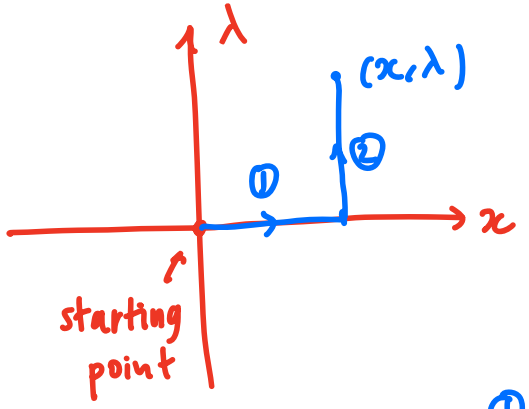
W_{m_1}: energy @ end point; W_{m_0}: energy @ start point

Key Assumption

* Coupling field is a lossless field (conservative field)
* What does having a conservative field mean? Energy change is path-independent.



Evaluating the Integrals



Assume:

- $x=0, \lambda=0$ - starting point
- $W_{m_0}=0$ (everything is turned off)
- $\lambda=0$ (no current) $\rightarrow f^e=0$

① Integrate along path ①:

$$W_m - W_{fp_0} = \int_0^0 i\,d\lambda - \int_0^x f^e\,d\hat{x} = 0$$

② Integrate along path ②

$$W_m = \int_0^{\lambda} i(\hat{\lambda}, x|_{\text{const}})\,d\hat{\lambda} - \int_x^x f\,d\hat{x}$$

$f \to 0$

⇓ Next, calc. the force: $f^e = -\left.\frac{\partial W_m(\lambda, x)}{\partial x}\right|_{\lambda=\text{const}}$

Example 3

- Let $\lambda = L(x)i$. What is the energy stored in the coupling field?

$\Rightarrow i = \frac{\lambda}{L(x)}$

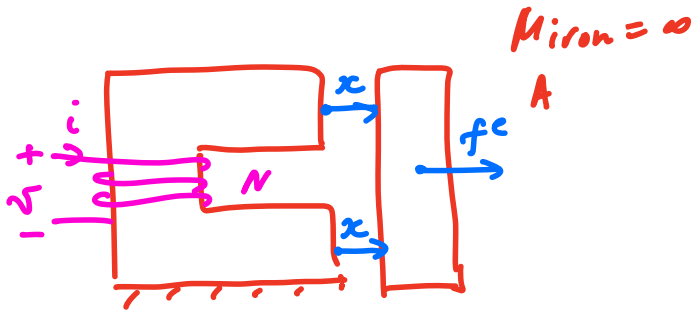
$\lambda = L(x)i \rightarrow$ linear

$$W_m = \int_0^{\lambda} i(\hat{\lambda}, x \mid \text{const})\, d\hat{\lambda} = \int_0^{\lambda} \frac{\hat{\lambda}}{L(x)}\, d\hat{\lambda} = \boxed{\frac{\lambda^2}{2L(x)} = W_m(\lambda, x)}$$

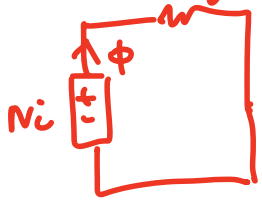
Always true for el. linear systems

Example 4

Q: compute $f^e(i,x)$ via $W_m(\lambda,x)$



① MEC



$$\Rightarrow \lambda = N\phi = \underbrace{\frac{N^2 \mu_0 A}{2x}}_{L(x)} \cdot i$$

el. linear

- $W_m(\lambda,x) = \int_0^{\lambda} i(\vec{\lambda}, x|_{const})\, d\vec{\lambda}$
- $f^e = -\frac{\partial W_m(\lambda,x)}{\partial x}\Big|_{\lambda = const}$

if el. linear

$$W_m = \frac{\lambda^2}{2L(x)}$$

② $W_m = \frac{1}{2}\left(\frac{2x}{\mu_0 A N^2}\right)\lambda^2$

substitute here!

③ $f^e = -\frac{\partial W_m(\lambda,x)}{\partial x}\Big|_{\lambda = const} = -\frac{1}{2}\left(\frac{2\lambda^2}{\mu_0 A N^2}\right) =$

$= \ominus \frac{1}{2}\left(\frac{\mu_0 A N^2}{2x^2}\right) i^2 \; [N]$

↑ f^e is in a direction opposite to x, attractive, wants to close the gap, minimize reluctance, energy stored

↑ f^e is always attractive irrespective of the polarity of i

Conservation of Power

- $\frac{dW_m}{dt} = i\frac{d\lambda}{dt} - f^e\frac{dx}{dt}$

$\frac{d(i\lambda)}{dt} = i\frac{d\lambda}{dt} + \lambda\frac{di}{dt}$

$i\frac{d\lambda}{dt} = \frac{d(i\lambda)}{dt} - \lambda\frac{di}{dt}$

- $\frac{dW_m}{dt} = \frac{d(i\lambda)}{dt} - \lambda\frac{di}{dt} - f^e\frac{dx}{dt}$

- $\frac{d\overbrace{(i\lambda - W_m)}^{=W_m'}}{dt} = \lambda\frac{di}{dt} + f^e\frac{dx}{dt}$

Define the Co-Energy

- $W_m' = i\lambda - W_m \Rightarrow \boxed{W_m + W_m' = i\lambda}$ Always true for linear and non-linear systems

- No physical meaning
- Mathematical construct to simplify calculations

$$\boxed{\frac{dW_m'}{dt} = \lambda \frac{di}{dt} + f^e \frac{dx}{dt}}$$

Note: $W_m'(i,x)$ (i and x as indep. variables)

Substitute Back into Conservation of Power

$$* \quad \frac{dW_m'}{dt} = \lambda \frac{di}{dt} + f^e \frac{dx}{dt}$$

$$* \quad \frac{dW_m'(i,x)}{dt} = \frac{\partial W_m'}{\partial i}\frac{di}{dt} + \frac{\partial W_m'}{\partial x}\frac{dx}{dt}$$

$$\lambda = \left.\frac{\partial W_m'(i,x)}{\partial i}\right|_{x=\text{const}}$$

$$f^e = \left.\frac{\partial W_m'(i,x)}{\partial x}\right|_{i=\text{const}}$$

no "-" !

Calculating the Co-energy

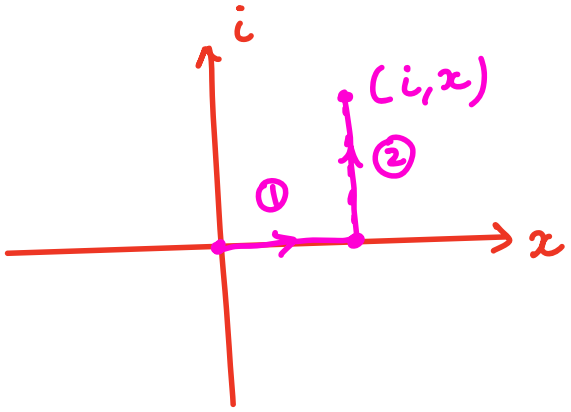
$$* \int_{t_0}^{t_1} \frac{dW_m'}{dt} dt = \int_{t_0}^{t_1} \lambda \frac{di}{dt} dt + \int_{t_0}^{t_1} f^e \frac{dx}{dt} dt$$

$$\int_{W_{m0}'}^{W_{m1}'} dW_m' = \int_{i_0}^{i_1} \lambda \, di + \int_{x_0}^{x_1} f^e dx$$

$$W_{m1}' - W_{m0}' = \int_{i_0}^{i_1} \lambda \, di + \int_{x_0}^{x_1} f^e dx$$

- Assume lossless field again

Evaluating the Integrals



Along ①:

$$W_m' - W_{m_0}' = \int_0^0 \lambda\, di + \int_0^x f^e\, dx = 0$$

($W_m' = W_{m_0}' = 0$; $\lambda = 0$; $f^e = 0$)

Along ②:

$$\boxed{W_m' = \int_0^i \lambda(\hat{i}, x|_{const})\, d\hat{i}} + \int_x^x f^e\, dx \quad (f^e = 0)$$

$$\Downarrow$$

Next, compute the force:

$$f^e = \left.\frac{\partial W_m'(i,x)}{\partial x}\right|_{i=const}$$

* $W_{m_0}' = 0$

* $i_0 = 0,\ f^e(i_0 = 0) = 0$

$$W_{m_1}' - W_{m_0}' = \int_{i_0}^{i_1} \lambda\, di + \int_{x_0}^{x_1} f^e\, dx$$

Example 5

- Let $\lambda = L(x)i$. What is the co-energy stored in the coupling field?

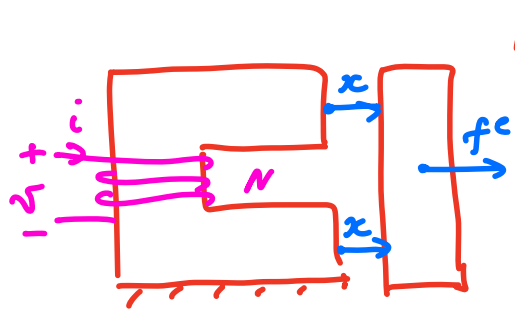
↳ el. linear

$$W_m'(i,x) = \int_0^i \lambda(\vec{i},x)\, d\vec{i} = \int_0^i L(x)\cdot\vec{i}\, d\vec{i} = \boxed{\frac{L(x)\cdot i^2}{2} = W_m'(i,x)}$$

Always true for el. linear systems

Example 6

Q: compute $f^e(i,x)$ via $W_m'(i,x)$

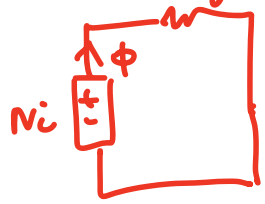


- $W_m'(i,x) = \int_0^i \lambda(\vec{i},x)\, d\vec{i}$
- $f^e = \left.\frac{\partial W_m'(i,x)}{\partial x}\right|_{i=\text{const}}$

if el. linear

$$W_m'(i,x) = \frac{L(x)\cdot i^2}{2}$$

① MEC



$$\Rightarrow \lambda = N\phi = \underbrace{\frac{N^2\mu_0 A}{2x}}_{L(x)} \cdot i \quad \text{el. linear}$$

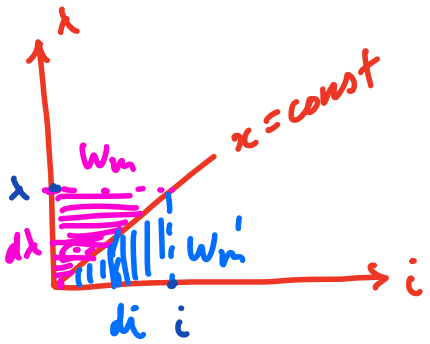
② $W_m'(i,x) = \frac{N^2\mu_0 A}{2x}\cdot\frac{i^2}{2}$

③ $f^e = -\frac{N^2\mu_0 A}{2x^2}\cdot\frac{i^2}{2}$ [N]

Energy vs. Co-energy

El. linear systems

$\lambda = L(x) \cdot i$



$$W_m = W_m'$$

numerically equal!

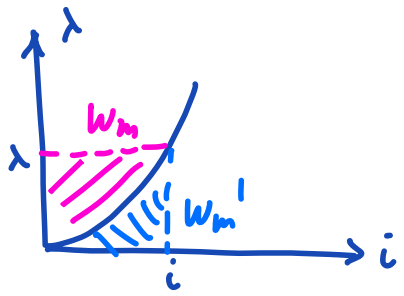
$$W_m = \int_0^{\lambda} i(\hat{\lambda}, x)\, d\hat{\lambda}$$

$$W_m' = \int_0^{i} \lambda(\hat{i}, x)\, d\hat{i}$$

$$W_m + W_m' = i\lambda$$

Non-linear systems

Ex: $\lambda = L(x) \cdot i^2$



$W_m \neq W_m'$

Ex: $\lambda = 2i^2$, $W_m(i, x) - ?$

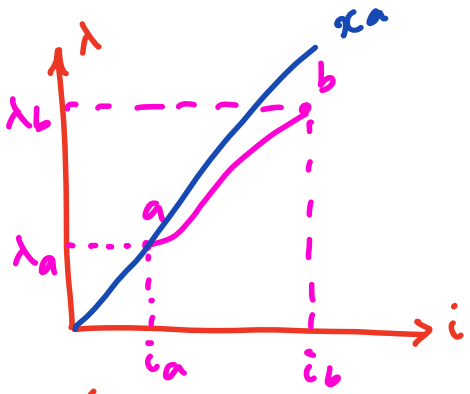
$\rightarrow W_m + W_m' = i\lambda \Rightarrow W_m = i\lambda - W_m'$

$\rightarrow W_m' = \int_0^{i} \lambda(\hat{i}, x)\, d\hat{i} = \frac{2i^3}{3}$ [J]

$\rightarrow W_m = i(2i^2) - \frac{2i^3}{3} = \frac{4}{3} \cdot i^3$ [J]

Energy conversion between two points

Q: How can we achieve continuous motion?



Given:

$\lambda = L(x) \cdot i$

Recall:

$$\Delta W_m = \int_{\lambda_a}^{\lambda_b} i \, d\lambda - \int_{x_a}^{x_b} f^e \, dx$$

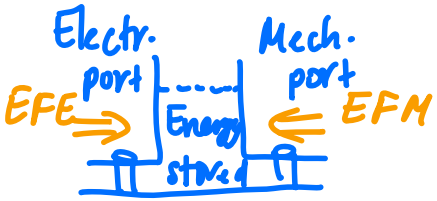
$$\Delta W_m = \int_{\lambda_a}^{\lambda_b} i \, d\lambda + \int_{x_a}^{x_b} -f^e \, dx$$

ΔW_m: change in stored energy is path-indep. due to conservative field

EFE $a \to b$: energy from electrical

EFM $a \to b$: energy from mechanical

Side note:

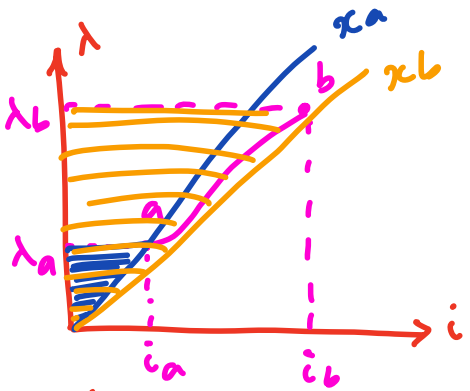


$$W_{mb} - W_{ma} = EFE_{a \to b} + EFM_{a \to b}$$

W_{mb}: energy @ endpoint

W_{ma}: energy @ start point

Example 7



Given:

$\lambda = L(x) \cdot i$

Q: EFM - ? $a \to 1$

$$W_{mb} - W_{ma} = \underset{a\to 1}{EFE} + \underset{a\to 1}{EFM}$$

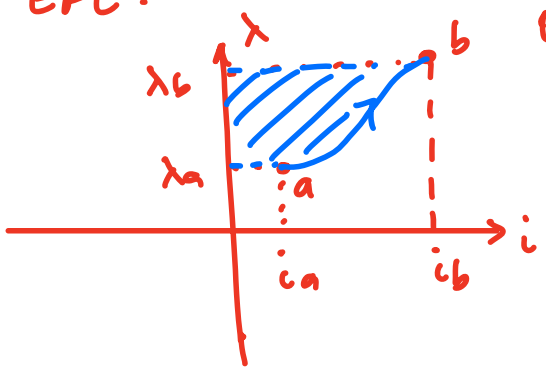
① $$W_m = \int_0^{\lambda} i(\hat{\lambda}, x)\, d\hat{\lambda} = \frac{\lambda^2}{2L(x)} = \frac{1}{2}\lambda i$$

$$W_{ma} = \frac{1}{2}\lambda_a i_a$$

$$W_{mb} = \frac{1}{2}\lambda_b i_b$$

Example 7-cont

EFE:



Q: EFM $_{a\to b}$ - ?

$$W_{mb} - W_{ma} = \underset{a\to b}{EFE} + \underset{a\to b}{EFM}$$

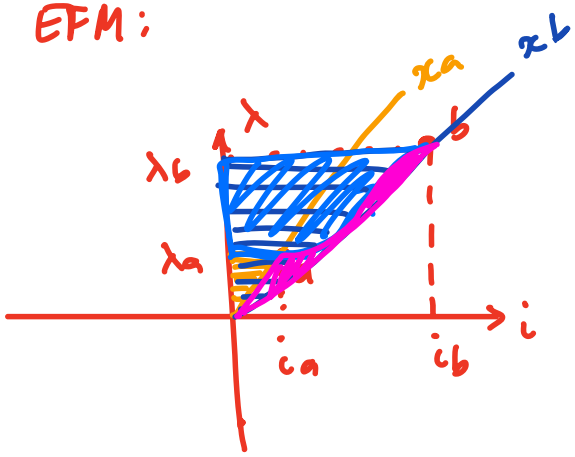
② Calculate EFE $_{a\to b}$:

$$\underset{a\to b}{EFE} = \int_{\lambda a}^{\lambda b} i\, d\lambda \quad \text{(integrate along the trajectory)} =$$

=

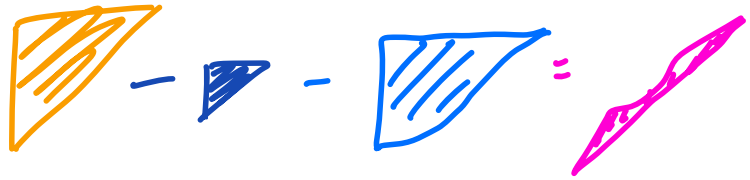


EFM:

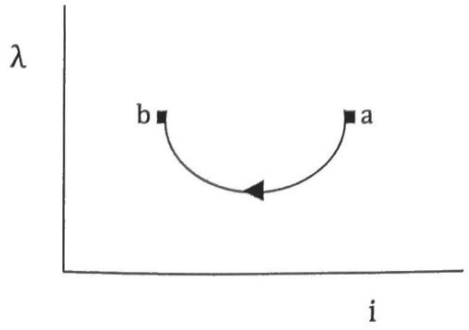


③ EFM $_{a\to b}$ = $W_{mb} - W_{ma}$ - EFE $_{a\to b}$ =

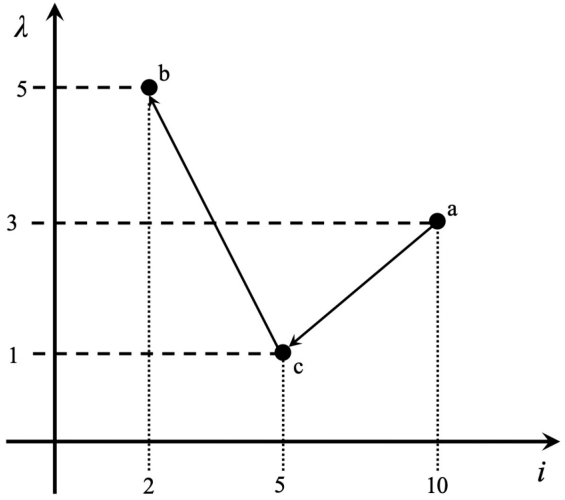
=



Example 8



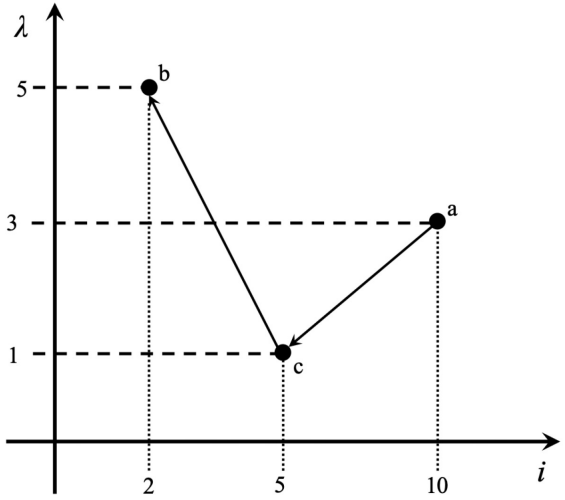
Example 9



An electromechanical system with $\lambda = L(x)i$ is moved from point a to b through point c as shown. The current is given as Amps and the flux linkage is given as Amp turns.

Energy over a cycle

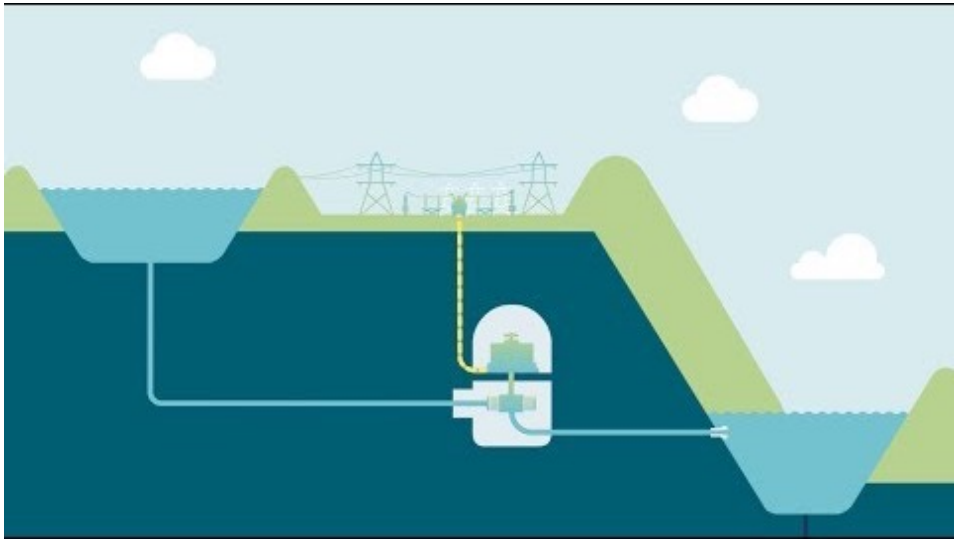
Example 9-cont



An electromechanical system with $\lambda = L(x)i$ is moved from point a to b through point c as shown.
The current is given as Amps and the flux linkage is given as Amp turns.

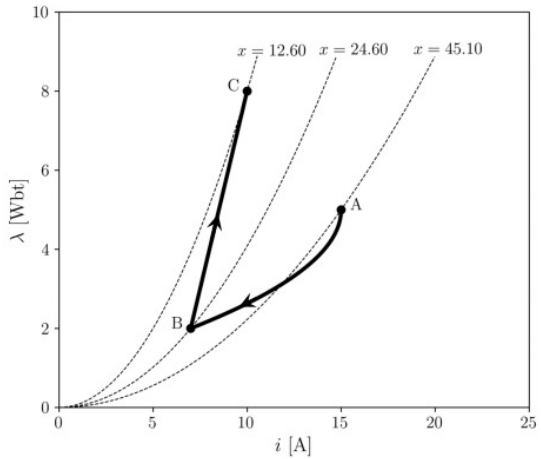
Flipping Direction

- Flipping the cycle direction converts:
  - Motor into generator
  - Generator into motor
  - Example: pumped hydro storage



https://www.youtube.com/watch?v=_PH0IJ-_qOI

Example 10



An electro-mechanical system that has the flux linkage relationship $\lambda = \frac{1}{x-0.1} i^2$ moves from point A ($i_a = 15$ A, $\lambda_a = 5$ Wbt) to point C ($i_c = 10$ A and $\lambda_c = 8$ Wbt) through the intermediate point B ($i_b = 7$ A, $\lambda_c = 2$ Wbt). The equation for the curve between points A and B is given as

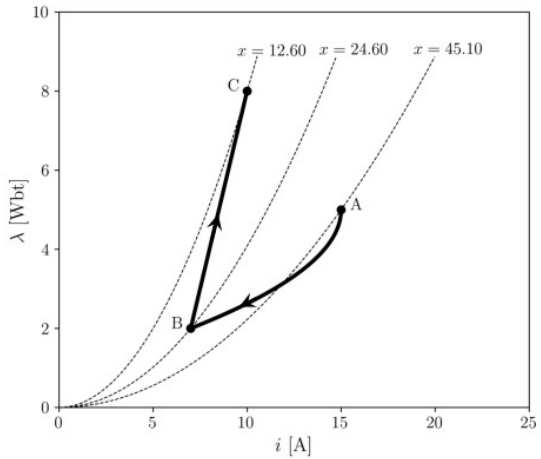
$$i = -\frac{8}{9}(\lambda - 5)^2 + 15$$

and the curve between points B and C is a straight line. The values for x at each given point are also given.

For the movement between these points, find:

a) The energy at point A and C.
b) The EFE when moving from point A to C.
c) The EFM when moving from point A to C.
d) If we move back from point C to point A along a straight line, what is the EFM over this cycle? Is the system behaving like a motor or a generator? Explain.

Example 10-cont



An electro-mechanical system that has the flux linkage relationship $\lambda = \frac{1}{x-0.1} i^2$ moves from point A ($i_a = 15$ A, $\lambda_a = 5$ Wbt) to point C ($i_c = 10$ A and $\lambda_c = 8$ Wbt) through the intermediate point B ($i_b = 7$ A, $\lambda_c = 2$ Wbt). The equation for the curve between points A and B is given as

$$i = -\frac{8}{9}(\lambda - 5)^2 + 15$$

and the curve between points B and C is a straight line. The values for x at each given point are also given.

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a) The energy at point A and C.
b) The EFE when moving from point A to C.
c) The EFM when moving from point A to C.
d) If we move back from point C to point A along a straight line, what is the EFM over this cycle? Is the system behaving like a motor or a generator? Explain.

Example 11

A certain system has flux linkage given as

$$\lambda = \frac{0.3}{x - 0.01} i^2 \qquad \text{with the constraint that } x > 0.01.$$

Find the energy from the mechanical system (EFM) and the energy from the electrical system (EFE) as the system moves from $x = 0.020$ m to $x = 0.015$ m while i is held constant at $i = 2$ A.

Example 11-cont

Multiple inputs system

Multiple inputs system

Example 12

Example 13